

Notes: Admati (Econometrica, 1985)

A Noisy Rational Expectations Equilibrium for Multi-Asset Securities Markets

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1 One-sentence summary

This note summarizes Admati (1985), which develops a closed-form noisy rational expectations equilibrium for a multi-asset securities market with heterogeneous private information, random supply noise, CARA-normal preferences, and a continuum of agents, and shows that cross-asset information effects can overturn several single-asset intuitions about prices, demands, and information aggregation.

2 Big picture

2.1 Question

The paper asks how equilibrium prices aggregate dispersed private information in a market with many risky assets, and how the information contained in prices changes agents' beliefs, demands, and comparative statics.

The key problem is that rational expectations prices perform two roles at the same time:

- they are **terms of trade**, because they enter the budget constraint and determine the cost of holding assets;
- they are also **signals**, because informed agents' demands move prices and therefore prices reveal information about payoffs.

In a single-asset model, these two effects are already subtle. In a multi-asset model, the problem becomes richer because the price of one asset may reveal information about other assets, and the supply shock of one asset can contaminate the informational content of the entire price vector.

2.2 Main result

The paper derives a closed-form linear noisy rational expectations equilibrium. In this equilibrium, each agent observes a private signal and the equilibrium price vector, forms a Gaussian posterior about the payoff vector, and chooses a linear demand function. Market clearing pins down the price vector as a linear function of the true payoff vector and the random supply vector.

The central equilibrium price takes the form

$$P = A_0 + A_1 F - A_2 Z, \tag{1}$$

where F is the vector of risky-asset payoffs and Z is the random supply vector. The matrices A_0, A_1, A_2 are solved explicitly from primitives.

2.3 Main economic message

The paper's main economic message is that once multiple risky assets interact, rational expectations prices no longer behave asset-by-asset. An asset's price can fall when its own payoff rises, a higher price can be interpreted as bad news for that asset, and an asset's demand can rise with its own price because the price increase changes beliefs in a favorable direction. These phenomena are not irrational; they arise from Bayesian inference in a correlated multi-asset environment.

3 Incremental contribution of the paper

3.1 Contribution relative to fully revealing rational expectations models

Earlier rational expectations models often focused on fully revealing equilibria, in which prices reveal all private information. That approach is conceptually powerful but creates well-known difficulties: if prices reveal all information, private information has no residual value, so agents may have no reason to acquire it in the first place.

Admati's contribution is to work with **noisy** rational expectations equilibria. Random supply noise prevents prices from perfectly revealing information. Agents can still learn from prices, but disagreement and private information remain relevant in equilibrium.

3.2 Contribution relative to single-asset noisy rational expectations models

Hellwig-style noisy rational expectations models had mainly been solved in the single-asset case. Admati extends the framework to many risky assets and obtains a closed-form solution under a continuum-agent limit.

The incremental advance is not merely technical. With many risky assets, cross-asset correlations create new economics:

- the price of one asset can convey information about other assets' payoffs and supplies;
- supply noise in one asset can prevent prices of other assets from becoming fully revealing;
- comparative statics can have signs that are impossible in one-asset models.

3.3 Contribution relative to CAPM-style asset pricing

The paper provides an asset-pricing model with heterogeneous information rather than homogeneous beliefs. In the CAPM, all investors share the same distribution and trade only because they differ in risk tolerance or endowments. Here, investors have different signals and use prices to update their beliefs. Therefore, the equilibrium cannot usually be summarized by a representative-agent CAPM.

3.4 Why the paper is foundational

This paper is foundational because it gives a tractable multi-asset laboratory for studying information aggregation, market efficiency, price informativeness, and portfolio demand under asymmetric information. It also provides building blocks for later applications such as insider trading, information acquisition, disclosure, trading volume, and performance evaluation.

4 Model setup

4.1 Assets and timing

There are two dates. Agents trade at date 0 and consume at date 1. There is one riskless asset and n risky assets.

The riskless asset pays R units of the consumption good. Risky asset i pays F_i , and the payoff vector is

$$F = (F_1, \dots, F_n)'. \quad (2)$$

Let P be the date-0 price vector of risky assets, and let D_a be agent a 's vector of risky-asset holdings. Taking the riskless asset as numeraire, final wealth is

$$W_{1a} = W_{0a}R + D_a'(F - RP). \quad (3)$$

Interpretation. The term $F - RP$ is the vector of excess payoffs relative to investing the price in the riskless asset. If agent a holds D_a , final wealth increases by $D_a'(F - RP)$.

4.2 Preferences

Each agent has constant absolute risk aversion utility

$$u_a(W) = -\exp\left(-\frac{W}{\rho_a}\right), \quad (4)$$

where $\rho_a > 0$ is the agent's risk tolerance. Larger ρ_a means the agent is less risk averse and reacts more strongly to expected excess returns.

CARA preferences are important because they remove wealth effects. This keeps demand linear under normality and allows the model to focus on information and risk-sharing effects.

4.3 Private signals

Agent a observes a private signal

$$Y_a = F + \varepsilon_a, \quad (5)$$

where ε_a is normally distributed with mean zero and variance-covariance matrix S_a . Error terms are independent across agents and independent of F and Z .

The precision of the signal is S_a^{-1} . A smaller S_a means a more precise signal. Agents can differ both in risk tolerance ρ_a and in signal precision S_a^{-1} .

4.4 Random supply noise

The per-capita supply of risky assets is random:

$$Z \sim N(\bar{Z}, U), \quad (6)$$

where U is positive definite in the main model.

Supply noise is the reason prices are not fully revealing. If prices move, agents cannot tell perfectly whether this reflects private information about F or a supply shock Z . This preserves a role for private information in equilibrium.

4.5 Distributional assumptions

The payoff vector is normally distributed,

$$F \sim N(\bar{F}, V), \quad (7)$$

where V is positive definite. The model allows general covariance structures for payoffs, supplies, and signal errors. This is where the multi-asset richness comes from: assets may have correlated payoffs, correlated supply shocks, and correlated signal errors.

5 Rational expectations equilibrium

5.1 Finite-agent equilibrium concept

In the finite economy, a rational expectations equilibrium consists of a price P and allocation functions $D_a(Y_a, P)$ such that

1. P is measurable with respect to all private signals and supply noise;
2. each agent maximizes expected utility conditional on both private information and the equilibrium price;
3. markets clear:

$$\frac{1}{N} \sum_{a=1}^N D_a(Y_a, P) = Z. \quad (8)$$

The rational expectations condition is the second condition: agents understand the equilibrium price function and correctly infer information from prices.

5.2 Why the paper moves to a continuum of agents

In a finite economy, each individual signal affects the price, and each agent must account for the covariance between her own signal error and the price. This creates technical difficulty and makes the agent both price-taking and price-relevant.

The continuum economy removes this issue. Individual signal errors wash out by a law of large numbers, so equilibrium prices depend only on aggregate payoff and supply variables, not on any individual error term.

5.3 Continuum-agent primitives

Agents are indexed by $a \in [0, 1]$. The economy is summarized by a measurable function

$$a \mapsto (\rho_a, S_a^{-1}). \quad (9)$$

Two aggregate objects play central roles:

$$\bar{\rho} = \int_0^1 \rho_a da, \quad (10)$$

which is average risk tolerance, and

$$Q = \int_0^1 \rho_a S_a^{-1} da, \quad (11)$$

which is the risk-tolerance-weighted aggregate precision of private information.

Economic meaning of Q . A signal from a more risk-tolerant investor moves equilibrium prices more because that investor trades more aggressively. Thus aggregate price informativeness is not the simple average of signal precision; it is precision weighted by trading aggressiveness.

6 Closed-form equilibrium solution

6.1 Linear price conjecture

The paper looks for a linear price function

$$P = A_0 + A_1 F - A_2 Z, \quad (12)$$

with A_2 nonsingular. Nonsingularity of A_2 means that supply noise affects prices in enough independent directions to prevent prices from being fully revealing.

6.2 Posterior beliefs

If the linear price conjecture holds, then (F, Y_a, P) is jointly normal. Therefore agent a 's posterior distribution of F conditional on (Y_a, P) is normal with mean

$$\mathbb{E}(F \mid Y_a, P) = B_{0a} + B_{1a} Y_a + B_{2a} P, \quad (13)$$

and variance-covariance matrix

$$V_a = \text{Var}(F \mid Y_a, P). \quad (14)$$

The posterior mean is linear in the private signal and price because of joint normality.

6.3 Demand function under CARA-normal preferences

Given posterior mean and variance, the CARA-normal demand is

$$D_a(Y_a, P) = \rho_a V_a^{-1} [B_{0a} + B_{1a} Y_a + (B_{2a} - RI)P]. \quad (15)$$

Derivation. The agent maximizes

$$\mathbb{E}_a \left[-\exp \left(-\frac{W_{0a} R + D'_a(F - RP)}{\rho_a} \right) \right]. \quad (16)$$

Under conditional normality, maximizing expected exponential utility is equivalent to maximizing

$$D'_a \mathbb{E}_a(F - RP) - \frac{1}{2\rho_a} D'_a V_a D_a. \quad (17)$$

The first-order condition is

$$\mathbb{E}_a(F - RP) - \frac{1}{\rho_a} V_a D_a = 0, \quad (18)$$

which gives

$$D_a = \rho_a V_a^{-1} \mathbb{E}_a(F - RP). \quad (19)$$

Substituting $\mathbb{E}_a(F) = B_{0a} + B_{1a}Y_a + B_{2a}P$ yields the expression above.

6.4 Market clearing and coefficient matching

Market clearing requires

$$\int_0^1 D_a(Y_a, P) da = Z. \quad (20)$$

Using $Y_a = F + \varepsilon_a$ and the continuum law of large numbers, the idiosyncratic errors average to zero. Therefore the aggregate demand is a linear function of F , P , and constants. Since $P = A_0 + A_1F - A_2Z$, matching coefficients yields the equilibrium matrices.

6.5 Key lemma: private-signal precision enters cleanly

The key technical step is the lemma

$$V_a^{-1} B_{1a} = S_a^{-1}. \quad (21)$$

Why this is important. It says that the marginal role of agent a 's private signal in her posterior, scaled by posterior precision, equals the signal's own precision. This result lets the aggregate private-information term collapse neatly to

$$\int_0^1 \rho_a V_a^{-1} B_{1a} da = \int_0^1 \rho_a S_a^{-1} da = Q. \quad (22)$$

6.6 Deriving the key lemma

Under the linear price conjecture, the covariance matrix of (F, Y_a, P) is

$$\begin{pmatrix} V & V & V A'_1 \\ V & V + S_a & V A'_1 \\ A_1 V & A_1 V & A_1 V A'_1 + A_2 U A'_2 \end{pmatrix}. \quad (23)$$

Normal regression implies that B_{1a} and B_{2a} satisfy

$$B_{1a}(V + S_a) + B_{2a}A_1V = V. \quad (24)$$

The conditional variance is

$$V_a = V - B_{1a}V - B_{2a}A_1V. \quad (25)$$

Adding the two equations gives

$$V_a = B_{1a}S_a. \quad (26)$$

Therefore

$$V_a^{-1}B_{1a} = S_a^{-1}. \quad (27)$$

Economic interpretation. Even though prices and private signals interact in a complicated way, the signal's direct contribution to posterior precision has the usual Bayesian form: precisions add.

7 The explicit equilibrium matrices

7.1 Useful notation

Define

$$M \equiv \bar{\rho}V^{-1} + \bar{\rho}QU^{-1}Q + Q. \quad (28)$$

This matrix summarizes the effective precision relevant for equilibrium pricing. It combines prior payoff precision, price-signal precision induced by supply noise, and aggregate private precision.

7.2 Equilibrium price coefficients

The unique linear noisy rational expectations equilibrium within the class $P = A_0 + A_1F - A_2Z$ with nonsingular A_2 has

$$A_0 = \frac{\bar{\rho}}{R}M^{-1} \left(V^{-1}\bar{F} + QU^{-1}\bar{Z} \right), \quad (29)$$

$$A_1 = \frac{1}{R}M^{-1} \left(Q + \bar{\rho}QU^{-1}Q \right), \quad (30)$$

$$A_2 = \frac{1}{R}M^{-1} \left(I + \bar{\rho}QU^{-1} \right). \quad (31)$$

Thus the price vector is

$$P = \frac{\bar{\rho}}{R}M^{-1} \left(V^{-1}\bar{F} + QU^{-1}\bar{Z} \right) + \frac{1}{R}M^{-1} \left(Q + \bar{\rho}QU^{-1}Q \right) F - \frac{1}{R}M^{-1} \left(I + \bar{\rho}QU^{-1} \right) Z. \quad (32)$$

7.3 Ex ante expected price

Taking expectations gives

$$\bar{P} \equiv \mathbb{E}(P) = \frac{1}{R} \left[\bar{F} - M^{-1}\bar{Z} \right]. \quad (33)$$

This result has the intuitive sign pattern at the ex ante mean level:

- a higher expected payoff raises expected price;
- a higher expected supply lowers expected price.

However, the paper's central point is that these intuitive signs need not hold for realized prices as functions of realized F and Z in a multi-asset equilibrium.

7.4 Posterior beliefs in equilibrium

Agent a 's posterior mean is

$$\mathbb{E}(F \mid Y_a, P) = B_{0a} + B_{1a}Y_a + B_{2a}P, \quad (34)$$

where

$$B_{1a} = V_a S_a^{-1}, \quad (35)$$

$$B_{2a} = R V_a Q U^{-1} A_2, \quad (36)$$

and

$$V_a = (V^{-1} + Q U^{-1} Q + S_a^{-1})^{-1}. \quad (37)$$

The posterior variance formula is the familiar precision-addition rule:

$$\text{posterior precision} = \text{prior precision} + \text{price-signal precision} + \text{private-signal precision}. \quad (38)$$

7.5 Equilibrium demand

The demand of agent a can be written as

$$D_a(Y_a, P) = G_{0a} + G_{1a}Y_a - G_{2a}P, \quad (39)$$

where

$$G_{1a} = \rho_a S_a^{-1}, \quad (40)$$

$$G_{2a} = \rho_a (K + R S_a^{-1}), \quad K = A_2^{-1} - R \bar{\rho} Q, \quad (41)$$

and G_{0a} is a constant term depending on primitives and ρ_a .

Important interpretation. The coefficient on the private signal is particularly simple: $G_{1a} = \rho_a S_a^{-1}$. A more risk-tolerant or better-informed agent trades more aggressively on private information.

8 How this differs from CAPM

8.1 Conditional CAPM intuition for each agent

Because each agent faces a conditionally normal return distribution and has CARA preferences, each agent's portfolio problem is mean-variance. Conditional on that agent's information, one can write a CAPM-like pricing relation with respect to that agent's conditional moments.

8.2 Why these conditional CAPMs do not aggregate

The conditional models do not aggregate into a single representative CAPM because different agents have different posterior means and covariance matrices. The market portfolio is generally not mean-variance efficient for any single common information set.

The reason is that conditional risk premia involve products of conditional moments. The law of iterated expectations does not turn products of conditional means and covariances into unconditional CAPM relationships.

8.3 Economic message

The model preserves the portfolio-demand logic of mean-variance optimization but rejects the homogeneous-beliefs aggregation behind the CAPM. Informed trading, heterogeneous signals, and price learning break the representative-investor simplification.

9 Equilibrium comparative statics and information effects

9.1 The positive stochastic-order result

The paper first proves a reassuring result. For each asset k , the conditional distribution of price P_k given payoff F_k is stochastically increasing in F_k . Similarly, the conditional distribution of F_k given P_k is stochastically increasing in P_k .

The reason is that

$$\text{Cov}(P_k, F_k) > 0. \quad (42)$$

Interpretation. If one only compares the pair (P_k, F_k) and lets all other variables be free, high prices are good news and high payoffs tend to be associated with high prices.

9.2 Why partial derivatives can have opposite signs

The stochastic-order result does not imply that the partial derivative $\partial P_k / \partial F_k$ must be positive when all other payoffs and supplies are held fixed.

In the price function

$$P = A_0 + A_1 F - A_2 Z, \quad (43)$$

$\partial P_k / \partial F_k$ is the k th diagonal element of A_1 . In a multi-asset model this element can be negative.

The difference is the same as the difference between simple regression and multiple regression. Pairwise covariance can be positive even when a partial regression coefficient is negative after conditioning on correlated variables.

9.3 Admati's two-asset example

The paper gives a two-asset example with

$$V = \begin{pmatrix} 1 & 5 \\ 5 & 26 \end{pmatrix}, \quad U = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}, \quad (44)$$

with $S_a = I$, $\bar{\rho} = R = 1$. The equilibrium price coefficients are

$$A_1 = A_2 = \begin{pmatrix} -0.02 & 0.19 \\ -0.45 & 1.08 \end{pmatrix}. \quad (45)$$

The posterior coefficients are

$$B_{1a} = V_a = \begin{pmatrix} 0.05 & 0.058 \\ 0.058 & 0.21 \end{pmatrix}, \quad B_{2a} = \begin{pmatrix} -0.24 & 0.17 \\ -2.42 & 1.24 \end{pmatrix}. \quad (46)$$

Demand coefficients are

$$G_{1a} = \rho_a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad G_{2a} = \rho_a \begin{pmatrix} 16.66 & -3 \\ 7 & -0.33 \end{pmatrix}. \quad (47)$$

9.4 What is counterintuitive in the example?

The example generates three striking results.

1. A higher price of asset 1, holding the price of asset 2 and the private signal fixed, lowers the expected payoff of asset 1.
2. The equilibrium price of asset 1 is decreasing in its own payoff and increasing in its own supply.
3. Asset 2 is a Giffen good: its demand is increasing in its own price.

None of these results would occur in a one-risky-asset model.

9.5 Economic intuition for bad news from a higher price

Suppose P_1 rises while P_2 stays fixed. A naive interpretation says that P_1 rose because informed traders received good news about asset 1. But in a correlated two-asset economy, this is incomplete.

If F_1 rises, then because F_1 and F_2 are strongly positively correlated, F_2 is also likely to rise, and by more. Then P_2 should also rise. If P_2 does not rise, agents infer that some offsetting supply shock likely occurred. Given the covariance structure of supplies, the supply shock that keeps P_2 fixed may also imply a large movement in the supply of asset 1. After sorting out all these correlated possibilities, an isolated increase in P_1 can be more consistent with worse payoff news than better payoff news.

9.6 Information effect and Giffen demand

In a standard Walrasian demand curve, a higher price reduces demand through the substitution effect. In a rational expectations equilibrium, a price change also changes beliefs.

This additional channel is the **information effect**:

$$\text{price increase} \Rightarrow \text{posterior belief changes} \Rightarrow \text{asset desirability changes}. \quad (48)$$

If the information effect is positive and strong enough, it can dominate the substitution effect. Then demand can rise when price rises. This is why a Giffen good can arise even without income effects.

10 Noisy equilibrium with some fixed supplies

10.1 Motivation

The main model assumes the supply noise covariance matrix U is full rank. Section 5 studies what happens when some asset supplies are fixed, so U is singular.

This question matters because fully revealing equilibria create conceptual difficulties. If no supplies are random, prices may reveal all payoff information. Then agents have no reason to acquire private information, creating a Grossman-Stiglitz type paradox.

10.2 Main insight

In a multi-asset model, uncertainty about the supply of one asset can prevent the entire price vector from being fully revealing. Even an asset with fixed supply may have a noisy price if its price is linked to other assets through payoff correlations, signal correlations, or portfolio substitution.

10.3 Economic intuition

Suppose asset 1 has fixed supply but asset 2 has random supply. If the two assets are correlated, the price of asset 1 depends not only on information about asset 1, but also on how agents substitute across assets and what they infer from the noisy price of asset 2. Thus noise in one market can contaminate the information content of other markets.

10.4 Implication

Very little supply noise may be enough to avoid fully revealing prices in a multi-asset setting. This makes noisy rational expectations equilibria more plausible than in a single-asset model, where each asset may require its own noise to prevent full revelation.

11 Extension: information about portfolios

11.1 General signal structure

The basic model assumes each signal directly concerns the payoff vector:

$$Y_a = F + \varepsilon_a. \quad (49)$$

The paper extends the model to signals of the form

$$Y_a = CF + \varepsilon_a, \quad (50)$$

where C is a $k \times n$ matrix. This allows agents to receive information about portfolios or linear combinations of assets, not necessarily about each asset separately.

11.2 Equilibrium with portfolio signals

In this generalized model, the solution has the same form after replacing S_a^{-1} with $C'S_a^{-1}C$ and replacing Q with the corresponding aggregate precision about the spanned payoff space.

The equilibrium remains linear:

$$P = A_0 + A_1F - A_2Z. \quad (51)$$

11.3 Indeterminacy and empirical equivalence

A key result is that different signal representations can be observationally equivalent if they span the same subspace of payoff information. For example, information about assets individually and information about certain portfolios may generate the same equilibrium price behavior if the spanned information space is the same.

This matters empirically because observed prices and returns may not identify the exact structure of private information.

11.4 Connection to timing and selectivity

The extension is useful for thinking about portfolio managers. Information about the market portfolio corresponds to market timing, while information about residual returns corresponds to selectivity. This distinction requires a multi-asset model; it cannot be properly studied in a one-risky-asset framework.

12 Extracted figures and tables from the paper

This paper is a theory paper and contains no standalone empirical tables or graphical figures. Therefore, there are no extracted figure/table images to place in this section. The main objects that would normally appear as extracted visuals are instead the closed-form equations, propositions, and derivations, which are typeset and explained directly in the sections above.

13 How to read the paper efficiently

13.1 First pass: understand the mechanism

On a first read, focus on the economic mechanism rather than every matrix manipulation:

- prices aggregate information;
- supply noise prevents full revelation;
- agents infer payoffs from both private signals and prices;
- in multiple assets, prices must be interpreted jointly, not separately.

13.2 Second pass: follow the closed-form solution

On a second read, focus on the sequence:

1. conjecture a linear price function;
2. compute Gaussian posterior beliefs;
3. derive CARA-normal demand;
4. impose market clearing;

5. use the key lemma $V_a^{-1}B_{1a} = S_a^{-1}$;
6. solve for A_0, A_1, A_2 .

13.3 Third pass: internalize Section 4

Section 4 is the conceptual heart of the paper. The important distinction is between:

- a pairwise statement: high P_k is good news about F_k on average;
- a partial-derivative statement: changing P_k while holding other prices fixed may be bad news.

This is the key to understanding the strange comparative statics.

14 Implications for finance research

14.1 Information aggregation

The paper shows that prices can aggregate information without fully revealing it. This makes it a natural framework for studying informational efficiency in markets where private information is valuable but partially impounded into prices.

14.2 Trading volume and disclosure

Because prices are noisy signals, disclosure policy, insider trading, and information acquisition can be analyzed in this framework. Random supply noise creates room for informed trading and costly information acquisition.

14.3 Performance evaluation

A multi-asset model is essential for evaluating portfolio managers. Skill may involve timing market-wide information or selecting specific assets. The distinction cannot be captured with one risky asset.

14.4 Asset-pricing anomalies under asymmetric information

The paper warns against interpreting demand curves or price reactions using single-asset intuition. A positive price movement may be bad news once other price movements are controlled for. This idea is useful for empirical work using multiple correlated assets, such as stocks, bonds, credit derivatives, or factor portfolios.

15 Limitations

15.1 CARA-normal structure

The closed-form solution depends heavily on CARA utility and joint normality. These assumptions make beliefs and demands linear, but they rule out wealth effects and may oversimplify tail risks.

15.2 Static two-period setup

The model is static. It does not describe dynamic learning, repeated trading, or how information decays over time. Extending the model to multiple periods would be important but technically difficult.

15.3 Continuum of agents

The continuum-agent economy creates tractability, but it also means the market as a whole may know payoffs perfectly. This produces some implementation issues when prices become fully revealing about certain portfolios.

15.4 Empirical identification

The extension shows that different private-information structures can be observationally equivalent if they span the same payoff space. This makes empirical identification of information structure challenging.

16 Conclusion

Admati (1985) provides a tractable closed-form noisy rational expectations equilibrium for a rich multi-asset market. The central technical achievement is the explicit solution for the linear equilibrium price, posterior beliefs, and demand functions. The central economic insight is that in multi-asset markets, price signals are interpreted jointly. Cross-asset correlations and supply noise can overturn single-asset intuitions: an asset's own price may be bad news, an asset's price may fall with its own payoff, and demand may rise with price because the information effect dominates the substitution effect.

The paper is therefore not only a technical contribution to rational expectations theory, but also a conceptual warning: in markets with multiple correlated securities, prices are equilibrium signals whose meaning depends on the full covariance structure of payoffs, supply shocks, and private information.